Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **Python/C++ systems** and **machine-learning pipelines** through simulation, multithreading, testing, debugging, and performance optimization.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Operating Systems, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Tripled throughput across 1K+ AI jobs by restructuring **concurrent backend orchestration** with explicit states, failure isolation, recoverable processing, and observable workflows under sustained production load reliably daily.
- Reduced dashboard latency from 1.2s to 350ms by optimizing **indexed SQL queries** and multi-tenant data models supporting 500+ users with normalized schemas and efficient access patterns reliably.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive, repeatable workflows during concurrent model execution and production debugging reliably.
- Maintained 90%+ grading consistency across 10K+ submissions through **structured evaluation workflows**, model switching, fallback paths, persistent results, and instructor review controls supporting reproducible verification reliably at scale.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Improved long-horizon trajectory planning by 18% in simulation by building a **PyTorch decision-transformer pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Raised closed-loop stability 15% through **quantitative verification** of HD maps, ego paths, nearby agents, attention, and distance-aware interactions across diverse autonomous-driving simulations and edge cases repeatedly and reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Automated builds and 20+ unit/integration suites through **Jenkins and GitHub**, adding failure reports and cross-platform validation across Linux, Windows, and macOS with repeatable nightly workflows reliably daily.
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments reliably at scale.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst - Research Fellowship *India*

- Achieved 96%+ classification accuracy across 650K+ URLs by comparing **Random Forest, CNN, RNN, LSTM, Bi-LSTM, and GRU models** within a reproducible machine-learning evaluation pipeline for phishing and malware detection.
- Engineered 21 behavioral indicators covering URL length, IP presence, and suspicious keywords, translating noisy inputs into **classification features** that exposed phishing and malware patterns reliably at scale.

Skills

Languages: Python, C++, Java, SQL, Bash, Shell, JavaScript
Machine Learning: PyTorch, Transformers, CNN, RNN, LSTM, GRU, Random Forest, Classification, Feature Engineering
Algorithms & Systems: Data Structures, Algorithms, Object-Oriented Design, Linux, Concurrency, Multithreading, Distributed Systems, System Design
Testing & Quality: Unit Testing, Integration Testing, Automated Testing, PyTest, Simulation, Debugging, Data Validation, Error Handling
Tooling & Delivery: Git, GitHub, Jenkins, CI/CD, Docker
Data & Engineering: PostgreSQL, Performance Optimization, Technical Documentation

Projects

Semiconductor Defect Inspection Pipeline - Design Concept

- Designed a **machine-vision inspection pipeline** concept using Python and PyTorch for image normalization, segmentation, defect classification, confidence calibration, structured evaluation, failure analysis, and reproducible experiment tracking workflows.

Cross-Subsystem Inspection Diagnostics - Design Concept

- Designed a **C++/Python diagnostic workflow** concept correlating image-acquisition events, motion states, hardware telemetry, and algorithm timings through bounded queues, trace identifiers, anomaly rules, and replayable test scenarios.

CaptionCraft - Image Caption Generator | Live

- Benchmarked BLIP, GIT, and ViT-GPT-2 across 10K images and 100K captions, improving **transformer inference accuracy** from 81% to 96% while building scalable real-time Flask APIs for caption generation.